Bayesian Framework for Canonical factorization of multiblock data

Abstract

The canonical factorization provides a common framework for multiblock methods: each block is decomposed as $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where $\mathbf{U}$ gathers components common to all the blocks and the diagonal matrix Σ_k measures the contribution of each component to block k. Two questions remain open in practice: the axes of the common subspace are only identified by conventions, and the number of dimensions has to be chosen. We impose orthonormal block loadings and show that the axes are then identified as soon as the profiles of the weights differ across blocks, a weaker condition than the one required by Multiple Co-Inertia Analysis (MCOA). Optimal MCOA (OMCOA), the simultaneous solution of this problem, is computed by alternating least squares. We then propose a Bayesian formulation whose full conditional distributions are matrix von Mises–Fisher, truncated normal and Gamma distributions: OMCOA is the iterated conditional modes algorithm of this posterior, and the Gibbs sampler provides credible intervals for the block weights. A spike-and-slab prior on the weights allows them to be exactly zero, which yields block-specific posterior inclusion probabilities and the posterior distribution of the dimension of the common subspace. A simulation study assesses the estimation of the weights and the selection of the dimensions as a function of the noise level, in relation to theoretical detectability thresholds, and compares the estimated dimension with the Bayesian singular value decomposition of Hoff and the Gavish–Donoho threshold.

Keywords: multiblock data analysis; canonical factorization; multiple co-inertia analysis; matrix von Mises–Fisher distribution; spike-and-slab prior; rank selection.

1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data commonly arise when several sets of variables are collected on the same individuals to capture different psychological domains, such as self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

Exploratory multiblock data analysis has four main objectives: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. A unified algebraic framework based on factorization strategies, called *canonical factorization*, was proposed by Bougeard et al. (2024) for multiblock methods to achieve these objectives. In practice, canonical factorization provides a structured framework for analyzing data variability at multiple levels: differences between and within blocks, and differences between observations through their projection onto a common subspace. It also facilitates the harmonization of the outputs of different methods and supports a criterion-based selection among competing approaches.

The canonical factorization is exact when the dimension of the common subspace equals the rank of the concatenated matrix, which can be large. Determining a reduced dimension, and which dimensions are active in each block, is therefore a major issue. Bayesian approaches address the first question for a single matrix: Hoff (2007) proposed a Bayesian singular value decomposition (SVD) that quantifies the uncertainty on the rank, and competing approaches were obtained with other priors on the singular values, such as the double exponential distribution (Zhai and Gutman, 2023), the repulsed normal distribution (Yuchi et al., 2023) and the horseshoe prior (Wang et al., 2026), which shrinks small singular values towards zero.

In the multiblock setting, the number of components and their activity in each block have also been addressed by rank-selection procedures for joint decompositions, such as AJIVE (Feng et al., 2018) and SLIDE (Gaynanova and Li, 2019), and by Bayesian factor models with sparsity-inducing priors on the loadings, such as Group Factor Analysis (Klami et al., 2015) and MOFA (Argelaguet et al., 2018), which switch off the factors that are inactive in a block. These approaches rely on unconstrained loadings; here, the orthonormality constraints of the canonical factorization identify the axes, and the activity of each dimension in each block is read directly on the block weights $\sigma_k^{(r)}$.

Following this line of work, we propose a Bayesian framework for the canonical factorization of multiblock data. The model imposes orthonormality constraints on both the common components and the block loadings, which makes the axes of the common subspace identifiable and leads to matrix von Mises–Fisher full conditional distributions. We derive an efficient Gibbs sampler and show that its conditional modes coincide with the updates of Optimal Multiple Co-Inertia Analysis (OMCOA), the simultaneous alternative to Multiple Co-Inertia Analysis (Chessel and Hanafi, 1996). We then propose a sparse version based on a spike-and-slab prior (Mitchell and Beauchamp, 1988; George and McCulloch, 1993) on the block weights, whose slab is scaled by the norm of each block. Unlike continuous shrinkage priors, and in the spirit of the point-mass prior used by Hoff (2007) for the singular values of a single matrix, this prior yields exact zeros and block-specific posterior inclusion probabilities, which directly provide an estimate of the dimension of the common subspace.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals,
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^K p_k)$ be the concatenation of these K-blocks,
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$,
- $\Sigma_k = \text{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$ are diagonal matrices,
- $\|.\|$ corresponds to the Frobenius norm,
- $\mathcal{V}_{N,R} = \{\mathbf{W} \in \mathcal{M}_{N,R}(\mathbb{R}) : \mathbf{W}^\intercal \mathbf{W} = \mathbf{I}_R\}$ denotes the Stiefel manifold.

Consider K column blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively in $\mathcal{M}_{N,p_1}(\mathbb{R}), ..., \mathcal{M}_{N,p_K}(\mathbb{R})$. Let $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^K p_k)$ be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

Bougeard et al. (2024) demonstrated that the $(K+1)-$blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = \text{rank}(\mathbf{X})$ as follows

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{U} \in \mathcal{V}_{N,R} \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \ldots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \ldots, K), \end{cases} \tag{1}$$

where

- $\mathbf{U} \in \mathcal{V}_{N,R}$ is the block-common components,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ are the block-loadings, and $\mathbf{V} = [\mathbf{V}_1|...|\mathbf{V}_K] \in \mathcal{M}_{P,R}(\mathbb{R})$,
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^K \Sigma_k^2$ that relate common and block-analyses.

Without loss of generality, we can assume $\sigma_k^{(r)} \geq 0$ for all k, r, since a joint sign change of $\sigma_k^{(r)}$ and $\mathbf{v}_k^{(r)}$ leaves the model and the constraints unchanged. Since the blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$ are column-centered, the block-common components are centered as well:

$$\mathbf{U} \in \mathcal{V}_{N,R}, \quad \mathbf{1}_N^\intercal\mathbf{U} = 0.$$

This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see Bougeard et al. (2024)). For all the multiblock methods, once $\mathbf{U}$ is fixed, the decomposition $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$ of each block is unique (up to sign). The methods differ in how the axes $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ themselves are determined. When the block loadings are only normalized ($\|\mathbf{v}_k^{(r)}\| = 1$), the optimal loadings are $\mathbf{v}_k^{(r)} = \mathbf{X}_k^\intercal\mathbf{u}^{(r)}/\|\mathbf{X}_k^\intercal\mathbf{u}^{(r)}\|$ and the least-squares criterion reduces to

$$\sum_{k=1}^K \left\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\right\|^2 = \|\mathbf{X}\|^2 - \left\|\mathbf{X}^\intercal\mathbf{U}\right\|^2,$$

which is invariant under any rotation $\mathbf{U} \mapsto \mathbf{U}\mathbf{O}$, $\mathbf{O}$ orthogonal: only span($\mathbf{U}$) is determined by the data, and the axes have to be fixed by an external convention, usually the SVD of $\mathbf{X}$. This convention becomes unstable when the singular values of $\mathbf{X}$ are close. When $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all k, the criterion is no longer rotation invariant, and the axes of $\mathbf{U}$ are identified (up to permutation and sign) by the block-specific profiles $\Sigma_1, \ldots, \Sigma_K$, as long as these profiles differ across blocks.

2.1 Canonical Factorisation under orthogonal block-loadings constraint

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^K \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for $k = 1, \ldots, K$, which requires $R \leq \min_k p_k$. Unlike (1), where the block loadings are only normalized, an exact factorization in dimension rank($\mathbf{X}$) is therefore

not attainable in general under this constraint: (2) defines a constrained version of the canonical factorization, whose benefit is the identifiability of the axes of $\mathbf{U}$ (Proposition 2). Chessel and Hanafi (1996) proposed a sequential solution, known as Multiple Co-Inertia Analysis (MCOA): the dimensions are computed one at a time, each from the blocks deflated with respect to the previous block loadings. This procedure yields orthonormal matrices $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$, and each stage is optimal in the least-squares sense given the previous ones. The resulting solution is nevertheless suboptimal with respect to (2), since each dimension is chosen without taking the next ones into account, so that the R dimensions are not fitted simultaneously. Optimal Multiple Co-Inertia Analysis (OMCOA) is proposed as a simultaneous solution of (2) under the constraints $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ $(k = 1, \ldots, K)$. It is computed by an alternating least squares (ALS) algorithm, in which each step minimizes (2) with respect to one set of parameters, the others being fixed:

- **Update of U:** $\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal$, where $\mathbf{X}\mathbf{H}^\intercal = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the SVD of $\mathbf{X}\mathbf{H}^\intercal$, with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$ (orthogonal Procrustes problem);
- **Update of $\mathbf{V}_k$:** $\mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$, for $k = 1, \ldots, K$;
- **Update of Σ_k:** $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$, for $k = 1, \ldots, K$.

Proposition 1. *The successive updates of $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ guarantee that the criterion defined in* (2) *decreases or remains constant at each step, so that the sequence of criterion values converges.*

The proof is given in the appendix.

Remark 1. *This guarantees the convergence of the criterion values, not of the parameters themselves. Since the limit may moreover be a local minimum, the algorithm is started from the MCOA solution and from several random initializations, and the solution with the lowest criterion is retained.*

Proposition 2 (Noiseless identifiability)**.** *Let $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, $k = 1, \ldots, K$, with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$, $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, $\sigma_k^{(r)} \geq 0$, and assume that every dimension is present in at least one block: for each r there is a k with $\sigma_k^{(r)} > 0$. Denote by $\mathbf{s}^{(r)} = (\sigma_1^{(r)}, \ldots, \sigma_K^{(r)})$ the profile of dimension r across the blocks and, for block weights $w_1, \ldots, w_K > 0$, by $\lambda_r(\mathbf{w}) = \sum_{k=1}^K w_k (\sigma_k^{(r)})^2$ the squared singular values of the weighted concatenated matrix $[\tilde{\mathbf{X}}_1 | \cdots | \tilde{\mathbf{X}}_K]$, $\tilde{\mathbf{X}}_k = \sqrt{w_k}\mathbf{X}_k$.*

(a) ***OMCOA.*** *If $\mathbf{s}^{(r)} \neq \mathbf{s}^{(r')}$ for all $r \neq r'$, every minimiser of* (2) *in dimension R, weighted or not, is equal to $(\mathbf{U}, \mathbf{V}_k, \Sigma_k)$ up to a common permutation of the dimensions and to the joint change of sign $(\mathbf{u}^{(r)}, \mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)}) \mapsto -(\mathbf{u}^{(r)}, \mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)})$; the columns $\mathbf{v}_k^{(r)}$ with $\sigma_k^{(r)} = 0$ are not determined.*

(b) ***MCOA.*** *If $\lambda_r(\mathbf{w}) \neq \lambda_{r'}(\mathbf{w})$ for all $r \neq r'$, MCOA applied to the weighted blocks $\tilde{\mathbf{X}}_k$ recovers $(\mathbf{U}, \mathbf{V}_k)$ up to the order and the signs of the dimensions, and the weights $\sqrt{w_k}\,\sigma_k^{(r)}$; its solution then coincides with the OMCOA solution (once the weights are divided by $\sqrt{w_k}$).*

(c) If $\lambda_r(\mathbf{w}) = \lambda_{r'}(\mathbf{w})$ for some $r \neq r'$, the MCOA solution is not determined by the data: at the first step where the largest remaining value is tied, any unit vector of the corresponding eigenspace is a solution, and the computed solution differs in general from $(\mathbf{U}, \mathbf{V}_k)$, even when the profiles are distinct.

The proof is given in the appendix.

Remark 2. *Equal profiles give equal values $\lambda_r(\mathbf{w})$ for every $\mathbf{w}$. Hence, whatever the weights, if the values $\lambda_r(\mathbf{w})$ are distinct, the profiles are distinct too, and MCOA and OMCOA recover the same solution. Conversely, for any weights there are designs with distinct profiles and tied values $\lambda_r(\mathbf{w})$, where only OMCOA is identified: OMCOA identifies strictly more configurations than MCOA, and its identifiability does not depend on the block weighting.*

3 Bayesian framework for Canonical Factorisation

A probabilistic counterpart of (2) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}(0, \tau_k^{-1}), \qquad k = 1, \dots, K, \tag{3}$$

where τ_k is the noise precision of block k. Since the blocks are centered, the latent components are constrained to be centered as well: $\mathbf{U} \in \mathcal{V}_{N,R}$ with $\mathbf{1}_N^\intercal\mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \dots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative von Mises–Fisher distribution). A random matrix $\mathbf{W} \in \mathcal{V}_{m,R}$ follows the matrix von Mises–Fisher distribution vMF($\mathbf{F}$), with parameter $\mathbf{F} \in \mathcal{M}_{m,R}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,R}$ is proportional to $\exp\{\mathrm{trace}(\mathbf{F}^\intercal\mathbf{W})\}$ (Hoff, 2009). If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the SVD of $\mathbf{F}$, the mode of vMF($\mathbf{F}$) is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.
- $\sigma_k^{(r)} \sim \mathcal{N}^+(0, \|\mathbf{X}_k\|^2/\lambda)$, a half-normal distribution scaled by the norm of the block, with a small dimensionless precision λ (e.g. $\lambda = 10^{-2}$);
- $p(\tau_k) \propto 1/\tau_k$, the Jeffreys prior (limit $a_0 = b_0 = 0$ of the Gamma prior).

Since $\sigma_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)} \leq \|\mathbf{X}_k\|$, the ratio $(\sigma_k^{(r)}/\|\mathbf{X}_k\|)^2$ is the proportion of the variance of block k explained by dimension r, and scaling the prior by $\|\mathbf{X}_k\|$ is equivalent to dividing each block by its Frobenius norm, the usual block weighting of multiblock methods. With the Jeffreys prior on τ_k, multiplying a block by a constant c multiplies $\sigma_k^{(r)}$ by c and τ_k by c^{-2} and leaves the posterior distribution of $\mathbf{U}$ and of the $\mathbf{V}_k$ unchanged: the model is applied to the raw (centred) blocks, and the weights are estimated on their original scale. The posterior distribution is proper as long as no block can be fitted exactly, which requires $R < \min_k p_k$: when $R = p_k$, $\mathbf{U}$ can span the column space of $\mathbf{X}_k$, the residual of block k can vanish, and the marginal posterior, proportional to $\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^{-(N-1)p_k}$ in this block, is not integrable.

3.1 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\left\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\right\|^2 = \|\mathbf{X}_k\|^2 - 2\,\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) + \sum_{r=1}^{R}(\sigma_k^{(r)})^2. \tag{4}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood is

$$\log p(\mathbf{X}_1, \dots, \mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K} \Big[\tfrac{(N-1)p_k}{2} \log \tau_k - \tfrac{\tau_k}{2} \big\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \big\|^2 \Big] + \text{const.}$$

By (4), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Proposition 3 (Full conditionals)**.** *Under model* (3) *and the above priors:*

1. ***Common space.*** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ with $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,R}$. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}}), \qquad \mathbf{F}_{\mathbf{U}} = \sum_{k=1}^{K} \tau_k \, \mathbf{X}_k \mathbf{V}_k \Sigma_k.$$

2. ***Block loadings.*** *For $k = 1, \dots, K$,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k}), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k \, \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k.$$

3. ***Block weights.*** *For $k = 1, \dots, K$ and $r = 1, \dots, R$, let $d_k^{(r)} = \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$. Then*

$$\sigma_k^{(r)} \mid \cdot \;\sim\; \mathcal{N}^+\Big(\frac{\tau_k \, d_k^{(r)}}{\tau_k + \lambda_k}, \; \frac{1}{\tau_k + \lambda_k} \Big), \qquad \lambda_k = \frac{\lambda}{\|\mathbf{X}_k\|^2},$$

   a normal distribution truncated to $(0, +\infty)$.

4. ***Noise precisions.*** *For $k = 1, \dots, K$,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(\frac{(N-1)p_k}{2}, \; \frac{1}{2} \big\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \big\|^2 \Big).$$

The modes of the full conditionals of Proposition 3 are

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_{\mathbf{U}} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k \mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k \mathbf{D}_k \mathbf{B}_k^\intercal, \qquad \sigma_k^{(r)} = \max\Big(0, \frac{\tau_k}{\tau_k + \lambda_k} \, d_k^{(r)}\Big).$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \cdots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1 \mathbf{V}_1^\intercal | \cdots | \Sigma_K \mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \operatorname{diag}(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)$, with the sign convention $\sigma_k^{(r)} \geq 0$ of Proposition 1. OMCOA is therefore the iterated conditional modes (ICM) algorithm (Besag, 1986) associated with this posterior, and the Gibbs sampler is its stochastic counterpart. With block-specific precisions, $\mathbf{F}_{\mathbf{U}} = \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion (2), a common precision $\tau_1 = \cdots = \tau_K = \tau$ can be used, with full conditional $\tau \mid \cdot \sim \mathrm{Gamma}\big(\sum_k (N-1)p_k/2, \; \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2/2\big)$.

3.2 Post-processing: label switching and signs

The posterior distribution is invariant under any permutation π of the R dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \ldots, \mathbf{V}_K$ and $\Sigma_1, \ldots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(r)}$ and $\mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\text{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\text{ref}}^\intercal \mathbf{U}$,

$$\hat{\pi} = \underset{\pi \in \mathfrak{S}_R}{\arg\max} \sum_{r=1}^{R} |C_{r,\pi(r)}|, \qquad s_r = \operatorname{sign}\big(C_{r,\hat{\pi}(r)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_r, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(r)}$ are then obtained from the empirical distribution of the aligned draws.

The resulting sampler is summarised in Algorithm 1.

Algorithm 1 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$; dimension $R < \min_k p_k$; number of iterations T; burn-in T_0; prior precision λ

Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \ldots, T$ **do**
  $\mathbf{F}_{\mathbf{U}} \leftarrow \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of vMF($\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}}$); $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \ldots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of vMF($\mathbf{F}_{\mathbf{V}_k}$)
    for $r = 1, \ldots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$; draw $\sigma_k^{(r)} \sim \mathcal{N}^+\big(\tau_k d_k^{(r)}/(\tau_k + \lambda_k),\, 1/(\tau_k + \lambda_k)\big)$, $\lambda_k = \lambda / \|\mathbf{X}_k\|^2$
    end for
    draw $\tau_k \sim \text{Gamma}\big((N-1)p_k/2,\ \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K, \tau_1, \ldots, \tau_K\big)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 3.2)
return the aligned draws

4 Sparse Bayesian Framework for canonical factorisation

In OMCOA, each block is approximated as $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where the diagonal matrix $\Sigma_k = \text{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$ measures how much each latent dimension $\mathbf{u}^{(r)}$ contributes to block k. OMCOA always returns strictly positive weights, so that this pattern has to be assessed afterwards. We propose a Bayesian version of OMCOA in which a spike-and-slab prior on the weights $\sigma_k^{(r)}$ allows them to be exactly zero. The dimension R is then a *maximal* rank: inactive dimensions are switched off by the sampler, and the effective rank is inferred from the data.

The sparsity of the model is based on the diagonal terms $\sigma_k^{(r)}$ as follows

$$\sigma_k^{(r)} = \gamma_k^{(r)}\, s_k^{(r)}, \qquad \gamma_k^{(r)} \in \{0, 1\}, \quad s_k^{(r)} > 0. \tag{5}$$

Priors. The difference between the two sets of prior distributions lies in the prior distributions of the diagonal terms:

- spike-and-slab on the weights, with a slab scaled by the norm of the block: $\gamma_k^{(r)} \mid \theta_r \sim \text{Bernoulli}(\theta_r)$ and $s_k^{(r)} \sim \mathcal{N}^+(0, \|\mathbf{X}_k\|^2/\lambda)$, i.e.

$$p(\sigma_k^{(r)} \mid \theta_r) = (1-\theta_r)\,\delta_0(\sigma_k^{(r)}) + \theta_r\,\mathcal{N}^+(\sigma_k^{(r)}; 0, \|\mathbf{X}_k\|^2/\lambda);$$

- $\theta_r \sim \text{Beta}(a_\theta, b_\theta)$, one inclusion probability per dimension, shared by the K blocks;
- $p(\tau_k) \propto 1/\tau_k$, as in Section 3.

As in Section 3, the slab is scaled by the block norm, so that λ is dimensionless, the selection of the dimensions does not depend on the scale of the blocks, and the posterior is proper when $R < \min_k p_k$. The slab precision λ is no longer small, however: it now controls the selection (see below). The prior depends on the data only through the block norms. Sharing θ_r across blocks borrows strength: a dimension that is active in several blocks is more likely to be active in another one, and a dimension that is inactive everywhere sees θ_r shrink towards zero.

As in the non-sparse case, the orthonormality of $\mathbf{U}$ and $\mathbf{V}_k$ gives

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_{r=1}^{R}\sigma_k^{(r)}d_k^{(r)} + \sum_{r=1}^{R}(\sigma_k^{(r)})^2, \qquad d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}. \tag{6}$$

Proposition 4 (Full conditionals)**.** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$.*

1. ***Common space:*** $\tilde{\mathbf{U}} \mid \cdot \sim \text{vMF}(\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})$ *with* $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$.
2. ***Block loadings:*** $\mathbf{V}_k \mid \cdot \sim \text{vMF}(\mathbf{F}_{\mathbf{V}_k})$ *with* $\mathbf{F}_{\mathbf{V}_k} = \tau_k\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. *When $\sigma_k^{(r)} = 0$, the r-th column of $\mathbf{F}_{\mathbf{V}_k}$ is zero and $\mathbf{v}_k^{(r)}$ is uniform on the orthogonal complement of the other columns.*
3. ***Weights (joint update of $\gamma_k^{(r)}$ and $\sigma_k^{(r)}$).*** *Let*

$$\lambda_k = \frac{\lambda}{\|\mathbf{X}_k\|^2}, \qquad \omega_k = \tau_k + \lambda_k, \qquad m_k^{(r)} = \frac{\tau_k\,d_k^{(r)}}{\omega_k}.$$

Then, with $\sigma_k^{(r)}$ integrated out,

$$P(\gamma_k^{(r)} = 1 \mid \cdot) = \frac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}, \qquad B_k^{(r)} = 2\sqrt{\frac{\lambda_k}{\omega_k}}\,\exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big)\,\Phi\Big(m_k^{(r)}\sqrt{\omega_k}\Big),$$

where Φ is the standard normal c.d.f.; and

$$\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \cdot \;\sim\; \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1}), \qquad \sigma_k^{(r)} \mid \gamma_k^{(r)} = 0, \cdot \;=\; 0.$$

4. ***Inclusion probabilities:*** $\theta_r \mid \cdot \sim \text{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\; b_\theta + K - \sum_k \gamma_k^{(r)}\big)$.
5. ***Noise precisions:*** $\tau_k \mid \cdot \sim \text{Gamma}\big((N-1)p_k/2,\; \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.

The matrix von Mises–Fisher distributions are sampled by one column-wise sweep per iteration, each column being drawn from a vector von Mises–Fisher distribution with Wood's algorithm (Wood, 1994), as in the non-sparse sampler. Updating $(\gamma_k^{(r)}, \sigma_k^{(r)})$ jointly, with $\sigma_k^{(r)}$ integrated out when drawing $\gamma_k^{(r)}$, is essential: drawing $\gamma_k^{(r)}$ given the current $\sigma_k^{(r)}$ would never allow the chain to leave $\sigma_k^{(r)} = 0$. The loading $\mathbf{v}_k^{(r)}$ is not integrated out, however: while $\gamma_k^{(r)} = 0$, it is drawn uniformly, so that $d_k^{(r)}$ is small and the reactivation of the dimension in block k may be slow. The mixing of the indicators should therefore be checked, for instance with chains started from $\gamma_k^{(r)} = 0$. Finally, the condition $R < \min_k p_k$ is also needed in practice with a proper Gamma prior on τ_k: when $R = p_k$, the dimensions can fit block k almost entirely, including its noise, so that τ_k is overestimated and the Bayes factors of this block are inflated, which keeps spurious dimensions active. In practice $\log B_k^{(r)}$ is computed on the log scale, with $\log \Phi$ evaluated directly, to avoid overflow. Moreover, $B_k^{(r)}$ contains the factor $\sqrt{\lambda_k/\omega_k}$: a very diffuse slab ($\lambda \to 0$) always favours the spike (Lindley's paradox). Since the slab is scaled by $\|\mathbf{X}_k\|$, the value of λ does not depend on the scale of the data: with $\lambda = 1$, the slab standard deviation is the norm of the block, so that a single dimension may explain most of its variance.

Remark 3 (Slab indexed on the noise)**.** *The slab may instead be indexed on the noise level of the block, $\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \tau_k \sim \mathcal{N}^+(0, g_k/\tau_k)$, in the spirit of Zellner's g-prior (Zellner, 1986; Liang et al., 2008), e.g. with $g_k = (N-1)p_k$, for which the slab scale is the expected Frobenius norm of the noise $\mathbf{E}_k$. The Bayes factor then becomes*

$$B_k^{(r)} = \frac{2}{\sqrt{1+g_k}} \exp\Big(\frac{g_k}{1+g_k}\,\frac{\tau_k (d_k^{(r)})^2}{2}\Big)\,\Phi\Big(d_k^{(r)}\sqrt{\frac{g_k\tau_k}{1+g_k}}\Big),$$

which only depends on the signal-to-noise ratio $\tau_k (d_k^{(r)})^2$, and the active weights enter the full conditional of τ_k, $\tau_k \mid \cdot \sim \mathrm{Gamma}\big(((N-1)p_k + \sum_r \gamma_k^{(r)})/2,\ \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2 + \sum_r (\sigma_k^{(r)})^2/(2g_k)\big)$. Both priors make the selection invariant to the scale of the blocks, but they differ in two respects. With the slab scaled by $\|\mathbf{X}_k\|$, the Bayes factor depends on $\tau_k\|\mathbf{X}_k\|^2$, that is on the whole signal of the block: a weak dimension is judged relative to the size of its block, and is penalised when the block also contains strong dimensions. With the slab indexed on the noise, each weight is judged against the noise only, independently of the other weights of the block, and the prior does not depend on the data; in return, the slab is coupled with τ_k, so that an overestimated noise precision inflates the Bayes factors. The two priors coincide when $g_k/\tau_k = \|\mathbf{X}_k\|^2/\lambda$, that is, for $\lambda = 1$, when $g_k = \tau_k\|\mathbf{X}_k\|^2 \approx (N-1)p_k + \tau_k\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$: with the default $g_k = (N-1)p_k$, they only differ when the total signal-to-noise ratio $\tau_k\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$ of the block is not negligible with respect to $(N-1)p_k$, i.e. at low noise levels, where the active weights are selected by both priors. We use the slab scaled by the block norm, which is consistent with the usual block weighting of OMCOA and keeps the interpretation of λ in terms of explained variance, and compare both priors in Section 5.

The posterior distribution of $\gamma_k^{(r)}$ gives a decision rule for the representation of block $\mathbf{X}_k$ on the component $\mathbf{u}^{(r)}$: the component $\mathbf{u}^{(r)}$ is declared active in block k if and only if

$$P\big(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \ldots, \mathbf{X}_K\big) > 1/2. \tag{7}$$

Moreover, this model makes it possible to estimate the effective dimension of the common subspace,

$$R_{\text{eff}} = \#\{r : \sum_k \gamma_k^{(r)} > 0\},$$

whose posterior distribution is estimated by its empirical distribution over the draws. The prior $\mathrm{Beta}(a_\theta, b_\theta)$ controls the expected sparsity; $a_\theta = b_\theta = 1$ is a neutral choice, and $b_\theta > a_\theta$ favours sparser structures.

The resulting sparse Gibbs sampler is summarised in Algorithm 2.

Algorithm 2 Sparse Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; maximal rank $R < \min_k p_k$; iterations T; burn-in T_0; slab precision λ; hyperparameters a_θ, b_θ

Initialization: $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with R dimensions on the blocks divided by their Frobenius norms, weights brought back to the original scale; $\gamma_k^{(r)} \leftarrow 1$; $\theta_r \leftarrow 1/2$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \dots, T$ **do**
  update $\tilde{\mathbf{U}}$ by one sweep of $\mathrm{vMF}\big(\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k\big)$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \dots, K$ **do**
    update $\mathbf{V}_k$ by one sweep of $\mathrm{vMF}\big(\tau_k \mathbf{X}_k^\intercal \mathbf{U}\Sigma_k\big)$
    for $r = 1, \dots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}$, $\omega_k \leftarrow \tau_k + \lambda/\|\mathbf{X}_k\|^2$, $m_k^{(r)} \leftarrow \tau_k d_k^{(r)}/\omega_k$
      draw $\gamma_k^{(r)} \sim \mathrm{Bernoulli}\Big(\dfrac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}\Big)$
      if $\gamma_k^{(r)} = 1$ **then** draw $\sigma_k^{(r)} \sim \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1})$ **else** $\sigma_k^{(r)} \leftarrow 0$
    end for
    draw $\tau_k \sim \mathrm{Gamma}\big((N-1)p_k/2,\ \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  for $r = 1, \dots, R$ **do**
    draw $\theta_r \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\ b_\theta + K - \sum_k \gamma_k^{(r)}\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \Sigma_1, \dots, \Sigma_K, \boldsymbol{\gamma}, \boldsymbol{\theta}, \boldsymbol{\tau}\big)$
  end if
end for
align the stored draws (permutation and signs) on a reference $\mathbf{U}_{\mathrm{ref}}$
return the aligned draws

5 Empirical performance via simulation

5.1 Simulation design

We consider $K = 3$ blocks with $p_1 = 10$, $p_2 = 5$, $p_3 = 15$ variables and $N = 100$ observations generated from

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad k = 1, 2, 3,$$

where $\mathbf{U} \in \mathcal{V}_{N,3}$ is column-centered, $\mathbf{V}_k \in \mathcal{V}_{p_k,3}$, Σ_k is the diagonal matrix of the block weights and $\mathbf{E}_k$ is a noise term. The true effective rank is $R_{\mathrm{eff}} = 3$. The diagonal matrices are set to

$$\Sigma_1 = \mathrm{diag}(0.7, 0.5, 0), \quad \Sigma_2 = \mathrm{diag}(0.5, 0.7, 0), \quad \Sigma_3 = \mathrm{diag}(0.3, 0, 0.5).$$

All the models are fitted with the maximal rank $R_{\max} = 4$, the largest value satisfying $R_{\max} < \min_k p_k$. For all models, the blocks were centered only, because Hoff's Bayesian SVD and the Gavish–Donoho threshold must be applied to the blocks without normalization.

Grid for noise variance: We assume that the terms $e_{\bullet\bullet}$ of the noise $\mathbf{E}_k$ are distributed from $\mathcal{N}(0, s^2)$. For a spectral method applied to a single block, an active weight w is asymptotically undetectable when the noise parameter s exceeds $w/(Np_k)^{1/4}$ (see Benaych-Georges and Nadakuditi, 2012, for details). Table 1 provides these thresholds as a function of the dimensional weights of the blocks. We set s within the interval from 0.02 to 0.18. Note that the first column of $\mathbf{U}$ should remain well estimated beyond $s = 0.05$, since it is shared by the three blocks, whereas the third column of $\mathbf{U}$ depends only on block 3 and should not be well estimated beyond $s = 0.08$.

dimension (r)	block (k)	$\sigma_k^{(r)}$	threshold
1	3	0.30	0.05
3	3	0.50	0.08
2	1	0.50	0.09
1	2	0.50	0.11
1	1	0.70	0.12
2	2	0.70	0.15

Table 1: Thresholds for the standard deviation of noise that make the weight of the dimension on the block undetectable.

[Sensitivity to the indexing of the slab: to be added, the sparse sampler with the slab scaled by the block norm ($\lambda = 1$) against the slab indexed on the noise ($g_k = (N-1)p_k$), on the same data sets, with $R_{\max} = 4$; e.g. selection probabilities as in Figure 3 and distribution of R_{eff} as in Figure 4.]

Slab parameter λ: The relative slab scale $s_0 = \lambda^{-1/2}$ gives the expected order of magnitude of a nonzero ratio $\sigma_k^{(r)}/\|\mathbf{X}_k\|$: under the slab, its mean is $s_0\sqrt{2/\pi}$. Since $\sigma_k^{(r)} \leq \|\mathbf{X}_k\|$, only the values of the ratio in $[0, 1]$ are attainable. With $\lambda = 1$, the slab puts 68% of its mass on $[0, 1]$ and allows a single dimension to explain most of the variance of a block. With $\lambda = 0.1$ ($s_0 \approx 3.2$), only 25% of its mass lies on attainable values, which penalizes the slab against the spike and makes the selection more conservative. With $\lambda = 10$ ($s_0 \approx 0.32$), the slab is concentrated on small weights, which favours the inclusion of weak dimensions and makes the selection less conservative.

The Gibbs samplers were run with $T = 6000$ iterations and a burn-in of $T_0 = 1000$. In order to evaluate the estimation of each dimension, all the fitted configurations $\hat{\mathbf{U}}$ were aligned on the true $\mathbf{U}$ (permutation and signs of the dimensions); in practice, the alignment is made on the OMCOA solution (Section 3.2). $B = 1000$ data sets were simulated in each case. For the effective rank, the estimate obtained using the Gibbs sparse sampler are compared with the estimate of the Bayesian SVD of Hoff on the concatenated block $\mathbf{X} = [\mathbf{X}_1|...|\mathbf{X}_K]$ (see Hoff, 2007, for details). The effective rank estimate is also compared to the rank obtained using the Gavish-Donoho threshold when the noise variance is unknown (see Gavish and Donoho, 2014, for details).

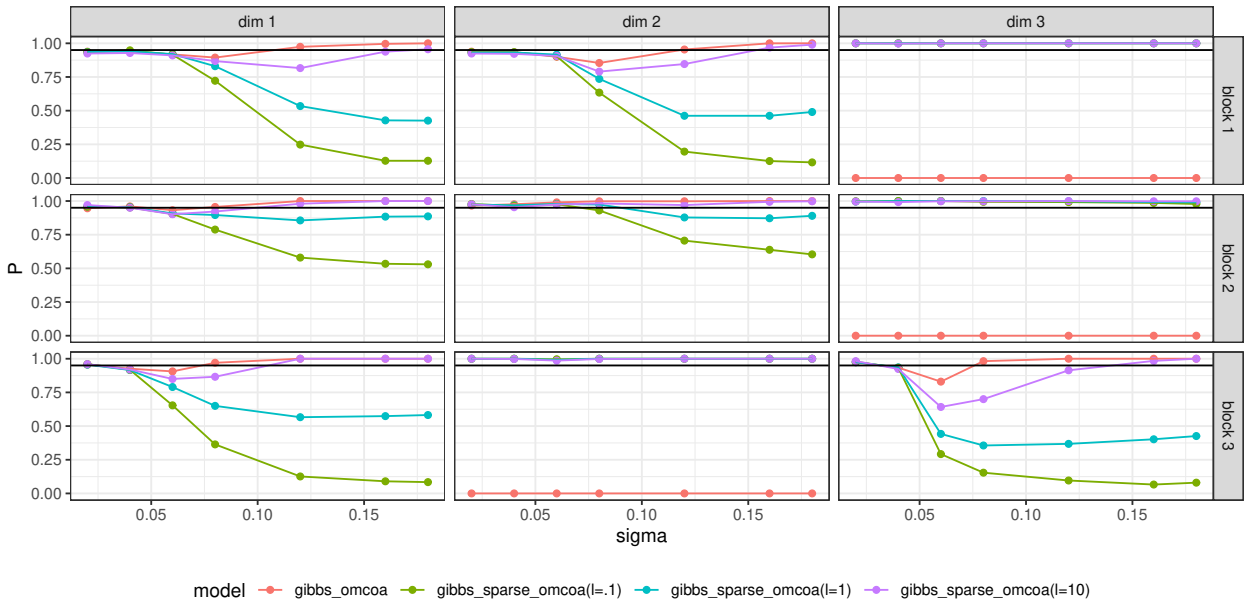


Figure 1: Empirical coverage probability of the weights $\sigma_k^{(r)}$ using gibbs-omcoa (in red), gibbs-sparse-omcoa (green for $\lambda = .1$, blue for $\lambda = 1$ and purple for $\lambda = 10$) as a function of the noise standard deviation s.

5.2 Results

Estimation of the weights $\sigma_k^{(r)}$:

Figure 1 provides the empirical coverage of the 95% credible intervals of the block weights for the active dimensions (dimensions 1 to 3), as a function of the noise standard deviation s and of the slab precision λ. The coverage was close to the nominal level for $s \leq 0.04$. For the weights equal to zero, the coverage of the non-sparse sampler was 0, since its half-normal prior excludes the value 0, whereas the sparse sampler covered the true value in all cases. For the active weights at high noise levels, the non-sparse sampler was over-conservative, with a coverage close to 1, since its credible intervals widened. For the active weights, the coverage of the sparse sampler decreased when s increased, because a weight that is not selected gets the degenerate interval $\{0\}$; as expected, this decrease was smaller for larger values of λ, which favour the inclusion. The weight of dimension 3, which is only present in block 3, was the first to be affected ($s > 0.05$), whereas the coverage of the weight of dimension 2 on block 2 remained good until $s = 0.08$. Figure 2 provides the estimated block weights of the active dimensions, with the 2.5% and 97.5% empirical quantiles over the B replications, as a function of s. When the weights were non-null, they were accurately estimated by the two Bayesian samplers and by MCOA and OMCOA for $s \leq 0.04$. The upward bias of MCOA and OMCOA increased with s, since these methods fit part of the noise, which inflates the estimated weights. The non-sparse Gibbs sampler degraded less. The sparse sampler behaved differently: a weight that is not detected is set to 0, so that the true zero weights were kept at 0, whereas the active weights were biased downwards at high noise levels; for instance, the weight 0.7 of dimension 1 on block 1 was estimated between about 0.1 and 0.3 for $s \geq 0.12$. This is a consequence of the selection, not an absence of bias.

For the inactive dimensions 4 and 5 (not shown), the estimated weights of MCOA, OMCOA and

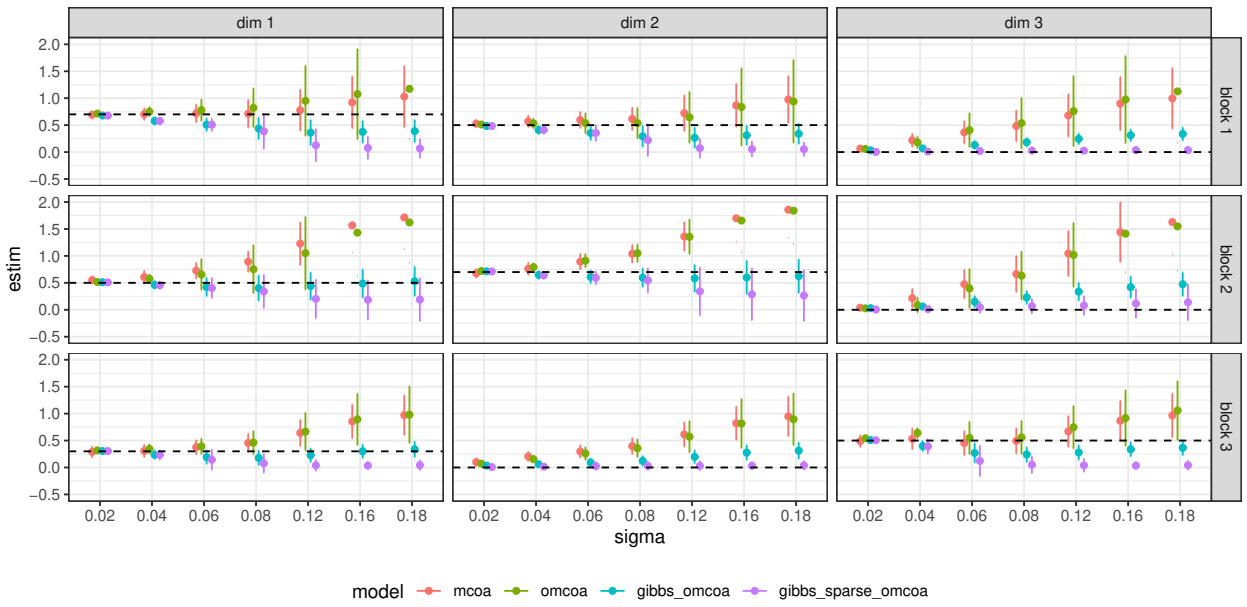


Figure 2: Estimated block weights, with the 2.5% and 97.5% empirical quantiles over the $B = 1000$ simulated data sets, for the three active dimensions (dimensions 1 through 3) obtained using the sequential approach (MCOA in red), the simultaneous approach (OMCOA in green), the Gibbs sampler for OMCOA (Gibbs OMCOA in light blue) and the sparse version (sparse Gibbs OMCOA with $\lambda = 1$ in purple), as a function of the noise standard deviation s. For Bayesian models, the estimate is equal to the posterior mean.

of the non-sparse sampler increased with s, since these methods fit the noise, whereas the sparse sampler kept them at 0 in almost all cases.

Analysis of sensitivity to the spike-and-slab parameters:

Slab parameter: We compare the performances in terms of the proportion of active dimensions that are selected using the decision rule defined in (7) as a function of the slab parameter λ (Figure 3). For all λ, when the noise standard deviation was low ($s < 0.05$), all the active dimensions were selected. When s increased, the active weights were lost in the following order: dimension 3 on block 3, dimension 1 on block 3, dimension 2 on block 1, dimension 1 on block 2, dimension 1 on block 1, and finally dimension 2 on block 2. Apart from the first two, this order follows the detectability thresholds of Table 1. The first two are inverted: the weight 0.3 of dimension 1 on block 3 (threshold 0.05) was still selected in about 65% of the data sets for $s = 0.06$ ($\lambda = 1$), whereas the weight 0.5 of dimension 3 on block 3 (threshold 0.08) was already selected in only about 25% of them. Dimension 1 is shared by the three blocks, so that $\mathbf{u}^{(1)}$ is well estimated from blocks 1 and 2 and its weight in block 3 remains detectable beyond the single-block threshold. Dimension 3 is only present in block 3: $\mathbf{u}^{(3)}$ is estimated from this block alone, and the Bayes factor requires a margin above the spectral threshold. The multiblock model therefore borrows strength across blocks and detects weights that a block-by-block analysis would miss. With the two more conservative values $\lambda = 0.1$ and $\lambda = 1$, dimensions 4 and 5, as well as the zero weights of dimensions 1 to 3 (dimension 3 on blocks 1 and 2, dimension 2 on block 3), were almost never selected. With $\lambda = 10$, the active weights were lost at slightly higher noise levels, at the cost of about 20% of false selections in block 2. A possible explanation is that $p_2 = R_{\max} = 5$: $\mathbf{V}_2$ is then a square orthogonal matrix, so that every direction of block 2 is associated with a dimension and the noise of this block always has a column to occupy. In this design, $\lambda = 1$ offers the best trade-off between the detection of weak weights and false selections.

Spike parameter:

Comparison of the estimates of the effective rank:

MCMC diagnostics: Bulk Effective Sample Size

sigma	$\hat{R}(R_{eff})$	Max Diff	$bulkESS(R_{eff})$	$\max(\hat{R}(\sigma_k^{(r)}))$	k	r	$\min(bulkESS(\sigma_k^{(r)}))$	k	r
0.04	1.00	0.04	488.00	1.01	1	2	203.00	3	4
0.06	1.05	0.11	99.00	1.03	2	4	78.00	2	4
0.08	1.01	0.07	131.00	1.02	1	4	97.00	2	4
0.12	1.00	0.06	100.00	1.01	1	2	223.00	3	3

[Paragraph to be rewritten once Hoff's SVD and the Gavish–Donoho threshold have been rerun on the non-standardised blocks.] Figure 4 provides the distribution of the estimated effective rank as a function of the noise standard deviation s, for the Gavish–Donoho criterion, the sparse Gibbs sampler and Hoff's Bayesian SVD. For values of s less than 0.06, the sparse Gibbs sampler identified three active dimensions in more than 50% of the simulated data, whereas the effective rank of 3, chosen according to the Hoff and Gavish-Donoho criteria, dropped very rapidly below 50% ($s > 0.025$). When no dimension was detectable ($s > 0.15$), Hoffs Bayesian SVD model overestimated

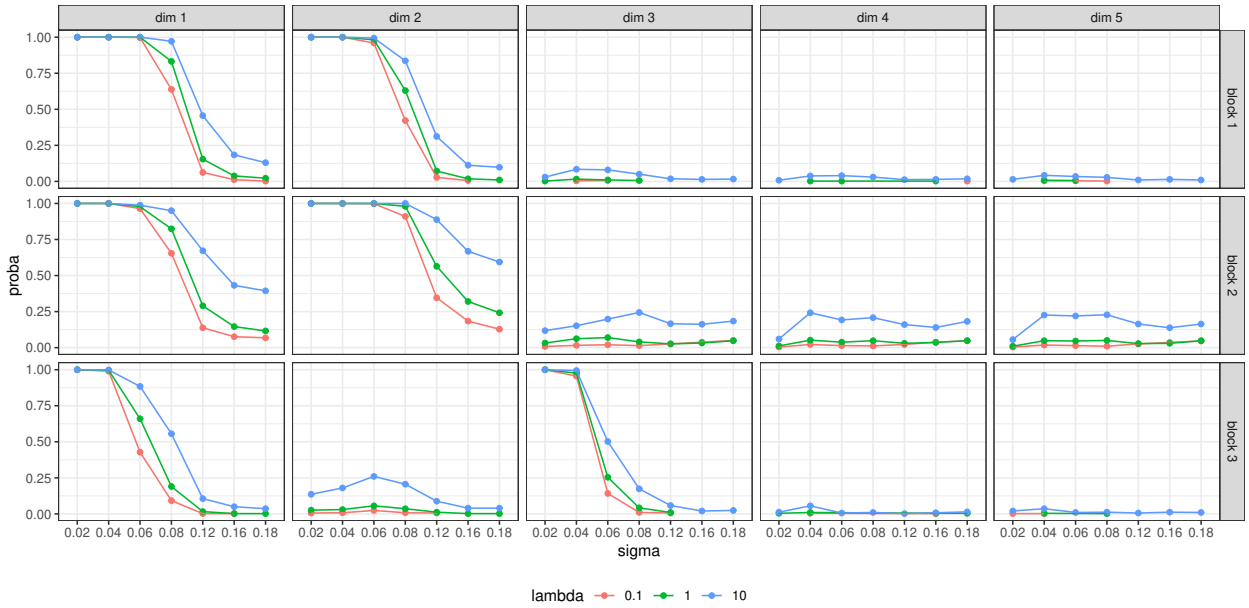


Figure 3: Empirical probability that a dimension is selected in a block, as a function of the noise standard deviation s and of the slab precision λ.

the effective rank because it retained five active dimensions in more than 75% of cases; the sparse Gibbs sampler found a rank between 1 and 3 most of the time; and the Gavish-Donoho criterion estimated the effective rank to be 1 most of the time.

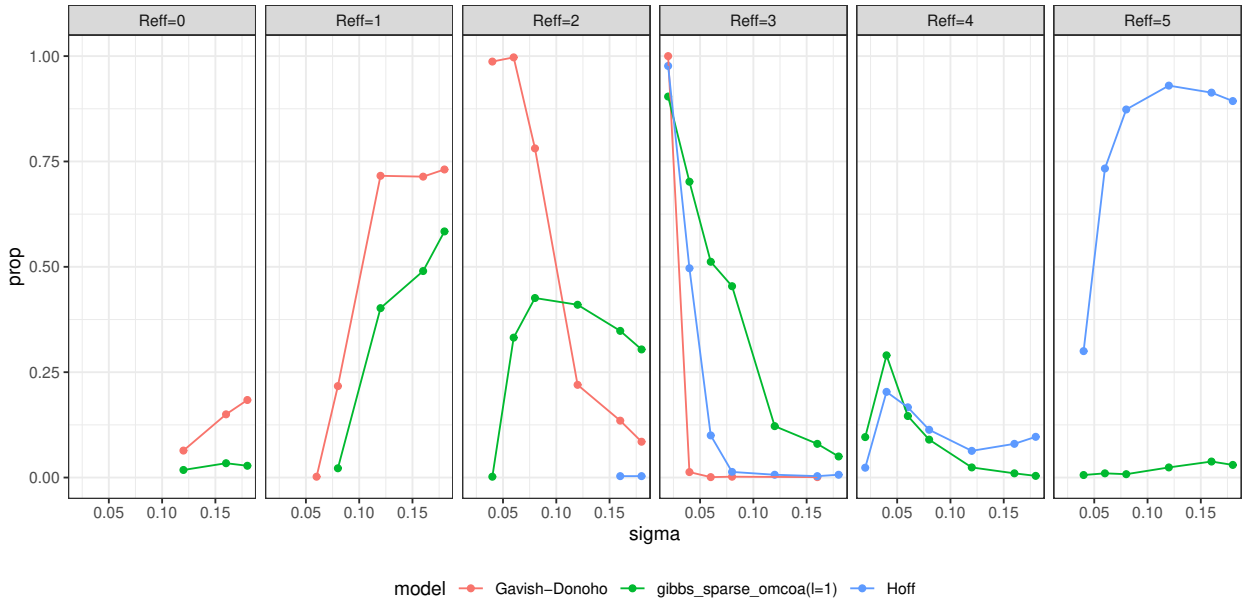


Figure 4: Distribution of the estimated effective rank with the Gavish–Donoho criterion, the sparse Gibbs sampler ($\lambda = 1$) and Hoff's Bayesian SVD, as a function of the noise standard deviation s.

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Appendix: proofs of the propositions

Proposition 1. The criterion to be minimized is:

$$\mathcal{L}(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K) = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\top\|^2,$$

subject to $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all $k = 1, \ldots, K$.

The algorithm is of the *Alternating Least Squares* (ALS) type: at each iteration, the criterion is minimized successively with respect to each parameter, while the others are held fixed. It therefore suffices to show that each individual update decreases or maintains $\mathcal{L}$.

Update of U With $\mathbf{V}_k$ and Σ_k fixed, let $\mathbf{H}_k = \Sigma_k \mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1 \mid \cdots \mid \mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. Then:

$$\mathcal{L} = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\mathbf{H}_k\|^2 = \|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2.$$

Expanding: $\|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2 = \|\mathbf{X}\|^2 - 2\ \mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) + \|\mathbf{U}\mathbf{H}^\intercal\|^2$.

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$: $\|\mathbf{U}\mathbf{H}^\intercal\|^2 = \mathrm{trace}(\mathbf{H}\mathbf{U}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}\mathbf{H}^\intercal) = \|\mathbf{H}\|^2$ (constant w.r.t. $\mathbf{U}$).

Minimizing $\mathcal{L}$ with respect to $\mathbf{U}$ is therefore equivalent to maximizing

$$\mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}^\intercal\mathbf{X}^\intercal\mathbf{U}) = \mathrm{trace}(\mathbf{U}^\intercal\mathbf{X}\mathbf{H}),$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{A}\mathbf{B}) = \mathrm{trace}(\mathbf{B}\mathbf{A})$ for $\mathbf{A}, \mathbf{B}$ of dimensions $N \times P$ and $P \times N$. The second equality comes from the equality of the trace of a square matrix and its transpose. By Cliff (1966), the maximum is attained at $\mathbf{U}^* = \mathbf{A}\mathbf{B}^\intercal$, where $\mathbf{X}\mathbf{H}^\intercal = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the thin SVD of $\mathbf{X}\mathbf{H}^\intercal \in \mathcal{M}_{N,R}(\mathbb{R})$. This update is the global minimizer of $\mathcal{L}$ with respect to $\mathbf{U}$, hence:

$$\mathcal{L}(\mathbf{U}^*, \ldots) \leq \mathcal{L}(\mathbf{U}_{\text{old}}, \ldots).$$

Update of $\mathbf{V}_k$ With $\mathbf{U}$ and Σ_k fixed, expanding the contribution of block k:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) + \|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2.$$

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$:

$$\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \mathrm{trace}(\mathbf{V}_k\Sigma_k^\intercal\mathbf{U}^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\Sigma_k^2) \quad \text{(constant w.r.t. } \mathbf{V}_k\text{)}.$$

Minimizing therefore reduces to maximizing $\mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k)$ subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$, which requires $R \le p_k$. By Cliff (1966), the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the thin SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. Note that Σ_k cannot be dropped: in general the polar factor of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ differs from that of $\mathbf{X}_k^\intercal\mathbf{U}$. If some $\sigma_k^{(r)}$ are zero, $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ is rank deficient and the maximizer is not unique, but any maximizer gives the same value of the criterion. Hence:

$$\mathcal{L}(\ldots, \mathbf{V}_k^*, \ldots) \le \mathcal{L}(\ldots, \mathbf{V}_{k,\mathrm{old}}, \ldots).$$

Update of Σ_k With $\mathbf{U}$ and $\mathbf{V}_k$ fixed (both having orthonormal columns), and $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, the contribution of block k is:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k) + \mathrm{trace}(\Sigma_k^2).$$

Since Σ_k is diagonal, it remains to minimize:

$$\sum_{r=1}^{R}\Big[(\sigma_k^{(r)})^2 - 2\,\sigma_k^{(r)}\,(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}\Big].$$

Each term is a convex parabola in $\sigma_k^{(r)}$, minimized at: $(\sigma_k^{(r)})^* = (\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}$, or, in matrix form:$\Sigma_k^* = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. This update is therefore the exact minimizer with respect to Σ_k. If $(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr} < 0$, replacing $\mathbf{v}_k^{(r)}$ by $-\mathbf{v}_k^{(r)}$ and $\sigma_k^{(r)}$ by $|(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}|$ keeps $\mathbf{V}_k$ orthonormal and leaves the criterion unchanged, so that the constraint $\sigma_k^{(r)} \ge 0$ costs nothing. Hence:

$$\mathcal{L}(\ldots, \Sigma_k^*, \ldots) \le \mathcal{L}(\ldots, \Sigma_{k,\mathrm{old}}, \ldots).$$

Global Convergence At each complete iteration (update of $\mathbf{U}$, then all $\mathbf{V}_k$, then all Σ_k), the criterion $\mathcal{L}$ is monotonically decreasing and bounded below by zero. A monotone sequence bounded below converges, so the sequence of values $(\mathcal{L}^{(t)})_{t\ge 0}$ converges:

$$\mathcal{L}^{(t+1)} \le \mathcal{L}^{(t)}, \quad \mathcal{L}^{(t)} \ge 0 \implies \mathcal{L}^{(t)} \xrightarrow[t\to\infty]{} \ell^* \ge 0.$$

□

Proposition 2. (a) The true parameters give the value 0 to criterion (2), so every minimiser $(\mathbf{U}', \mathbf{V}_k', \Sigma_k')$ satisfies $\mathbf{X}_k = \mathbf{U}'\Sigma_k'\mathbf{V}_k'^\intercal$ for all k; this does not depend on the weights given to the blocks. Since every dimension is present in some block, the column space of $\mathbf{X} = [\mathbf{X}_1|\cdots|\mathbf{X}_K]$ is the span of $\mathbf{U}$, of dimension R. It is contained in the span of $\mathbf{U}'$, which also has dimension R; hence $\mathbf{U}' = \mathbf{U}\mathbf{O}$ with $\mathbf{O}$ orthogonal. From $\mathbf{X}_k\mathbf{X}_k^\intercal = \mathbf{U}\Sigma_k^2\mathbf{U}^\intercal = \mathbf{U}'\Sigma_k'^2\mathbf{U}'^\intercal$ we get

$$\mathbf{O}^\intercal\Sigma_k^2\,\mathbf{O} = \Sigma_k'^2, \qquad k = 1, \ldots, K.$$

For $\mathbf{c} \in \mathbb{R}^K$, the diagonal matrix $\mathbf{A}(\mathbf{c}) = \sum_k c_k \Sigma_k^2$ has entries $a_r = \sum_k c_k (\sigma_k^{(r)})^2$, and $\mathbf{O}^\intercal \mathbf{A}(\mathbf{c}) \mathbf{O}$ is diagonal. Since the profiles are distinct and nonnegative, the squared profiles are distinct, and $\mathbf{c}$ can be chosen outside the finite union of hyperplanes $\{a_r = a_{r'}\}$, so that the entries of $\mathbf{A}(\mathbf{c})$ are pairwise distinct. An orthogonal matrix that diagonalises a diagonal matrix with distinct entries is a signed permutation matrix: $\mathbf{O} = \mathbf{P}\mathbf{D}$, with $\mathbf{P}$ a permutation matrix and $\mathbf{D} = \mathrm{diag}(\pm 1)$. Hence $\mathbf{U}' = \mathbf{U}\mathbf{P}\mathbf{D}$ and $\Sigma_k'^2 = \mathbf{P}^\intercal \Sigma_k^2 \mathbf{P}$, that is $\Sigma_k' = \mathbf{P}^\intercal \Sigma_k \mathbf{P}$ since the weights are nonnegative. Finally, $\mathbf{U}'^\intercal \mathbf{X}_k = \Sigma_k' \mathbf{V}_k'^\intercal$ gives, for every (k, r) with $\sigma_k'^{(r)} > 0$, $\mathbf{v}_k'^{(r)} = \mathbf{X}_k^\intercal \mathbf{u}'^{(r)} / \sigma_k'^{(r)}$, which is the matching column of $\mathbf{V}_k$ with the sign of $\mathbf{u}'^{(r)}$; the columns with $\sigma_k'^{(r)} = 0$ are only constrained by $\mathbf{V}_k' \in \mathcal{V}_{p_k, R}$.

(b) Write $\mathcal{R}_t$ for the set of dimensions not yet extracted at step t of MCOA ($\mathcal{R}_1 = \{1, \dots, R\}$), and assume that the deflated blocks are $\mathbf{X}_k^{(t)} = \sqrt{w_k} \sum_{r \in \mathcal{R}_t} \sigma_k^{(r)} \mathbf{u}^{(r)} \mathbf{v}_k^{(r)\intercal}$, which holds for $t = 1$. Then

$$\mathbf{X}^{(t)} \mathbf{X}^{(t)\intercal} = \sum_k \mathbf{X}_k^{(t)} \mathbf{X}_k^{(t)\intercal} = \sum_{r \in \mathcal{R}_t} \lambda_r(\mathbf{w})\, \mathbf{u}^{(r)} \mathbf{u}^{(r)\intercal},$$

because the columns of each $\mathbf{V}_k$ are orthonormal. The values $\lambda_r(\mathbf{w})$ being distinct, the leading left singular vector of $\mathbf{X}^{(t)}$ is $\pm \mathbf{u}^{(r_t)}$, with $r_t = \arg\max_{r \in \mathcal{R}_t} \lambda_r(\mathbf{w})$. Then $\mathbf{X}_k^{(t)\intercal} \mathbf{u}^{(r_t)} = \sqrt{w_k}\, \sigma_k^{(r_t)} \mathbf{v}_k^{(r_t)}$: the MCOA weight is $\sqrt{w_k}\, \sigma_k^{(r_t)}$ and, when it is positive, the normalised loading is $\pm \mathbf{v}_k^{(r_t)}$ (with the sign of $\mathbf{u}$), so that the deflation $\mathbf{X}_k^{(t)} (\mathbf{I} - \mathbf{v}_k^{(r_t)} \mathbf{v}_k^{(r_t)\intercal})$ removes exactly the term r_t. When the weight is zero, the term r_t is absent from block k and the block is left unchanged. The induction hypothesis holds at step $t+1$ with $\mathcal{R}_{t+1} = \mathcal{R}_t \setminus \{r_t\}$. After R steps, MCOA has recovered $\mathbf{U}$ and the $\mathbf{V}_k$ (columns with a positive weight) up to order and signs, and gives a zero value to criterion (2) for the weighted blocks: by (a), it coincides with the OMCOA solution.

(c) Let t be the first step where $\max_{r \in \mathcal{R}_t} \lambda_r(\mathbf{w})$ is attained by several dimensions $r \in \mathcal{T} \subset \mathcal{R}_t$. Up to step $t-1$, the argument of (b) applies, and the leading eigenvalue of $\mathbf{X}^{(t)} \mathbf{X}^{(t)\intercal}$ has the eigenspace $\mathrm{span}\{\mathbf{u}^{(r)}, r \in \mathcal{T}\}$, of dimension $|\mathcal{T}| \geq 2$. Every unit vector of this space maximises the MCOA criterion at step t; a vector other than $\pm \mathbf{u}^{(r)}$, $r \in \mathcal{T}$, gives a common component that is not a column of $\mathbf{U}$ up to sign, hence, by (a), a solution different from the OMCOA one. □

Proposition 3. (1)–(2): by (4), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k \,\mathrm{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})\} = \exp\{\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{U})\}$, and $\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{Q} \tilde{\mathbf{U}}) = \mathrm{trace}\big((\mathbf{Q}^\intercal \mathbf{F}_\mathbf{U})^\intercal \tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\mathrm{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}) = \mathrm{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal \mathbf{V}_k)/\tau_k$. (3): the terms of the log-likelihood involving $\sigma_k^{(r)}$ are, by (4), $-\frac{\tau_k}{2}[(\sigma_k^{(r)})^2 - 2\sigma_k^{(r)} d_k^{(r)}]$; adding the log-prior $-\frac{\lambda_k}{2}(\sigma_k^{(r)})^2$, $\lambda_k = \lambda / \|\mathbf{X}_k\|^2$, and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(r)} > 0$. (4): the likelihood contributes $\tau_k^{(N-1)p_k/2} \exp(-\tau_k \|\mathbf{X}_k - \mathbf{U} \Sigma_k \mathbf{V}_k^\intercal\|^2/2)$ and the Jeffreys prior τ_k^{-1}, which gives a Gamma kernel. □

Proposition 4. Items 1, 2, 4 and 5 follow as in the non-sparse sampler (linearity of (4) in $\mathbf{U}$ and $\mathbf{V}_k$, and conjugacy; for item 5, the Jeffreys prior contributes the factor τ_k^{-1}). For item 3, the terms of the log-likelihood involving $\sigma_k^{(r)}$ are, by (6), $\tau_k d_k^{(r)} \sigma_k^{(r)} - \frac{\tau_k}{2}(\sigma_k^{(r)})^2$. Under $\gamma_k^{(r)} = 0$, $\sigma_k^{(r)} = 0$ and this quantity is 0. Under $\gamma_k^{(r)} = 1$, integrating against the half-normal slab of density $2\sqrt{\lambda_k/(2\pi)} \exp(-\lambda_k s^2/2)$ on $(0, \infty)$, $\lambda_k = \lambda / \|\mathbf{X}_k\|^2$, gives

$$2\sqrt{\frac{\lambda_k}{2\pi}} \int_0^\infty \exp\Big(-\frac{\omega_k}{2} s^2 + \tau_k d_k^{(r)} s\Big) ds = 2\sqrt{\frac{\lambda_k}{\omega_k}} \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \Phi\big(m_k^{(r)} \sqrt{\omega_k}\big) = B_k^{(r)},$$

which is the Bayes factor of $\gamma_k^{(r)} = 1$ against $\gamma_k^{(r)} = 0$. The conditional of $\sigma_k^{(r)}$ given $\gamma_k^{(r)} = 1$ is obtained by completing the square. □

MCMC diagnostics F

